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 Calculator

Problem 5.1 Find the general solution $(x(t), y(t), z(t))$ to the system

$$\begin{aligned}\dot{x} &= 2x \\ \dot{y} &= -6x + 8y + 3z \\ \dot{z} &= 18x - 18y - 7z.\end{aligned}$$

Solution:

In matrix form, this is $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$, where $\mathbf{A} = \begin{pmatrix} 2 & 0 & 0 \\ -6 & 8 & 3 \\ 18 & -18 & -7 \end{pmatrix}$.

Step 1. Find the eigenvalues. To do this, compute

$$\det(\lambda \mathbf{I} - \mathbf{A}) = \begin{vmatrix} \lambda - 2 & 0 & 0 \\ 6 & \lambda - 8 & -3 \\ -18 & 18 & \lambda + 7 \end{vmatrix}.$$

Use Laplace expansion along the first row to get

$$(\lambda - 2) ((\lambda - 8)(\lambda + 7) - 18(-3)) = (\lambda - 2)(\lambda^2 - \lambda - 2) = (\lambda - 2)(\lambda - 2)(\lambda + 1),$$

so the eigenvalues are $2, 2, -1$.

Step 2. Find a basis of each eigenspace and write down the exponential solutions.

Eigenspace of $\lambda = 2$: This is the nullspace of

$$2\mathbf{I} - \mathbf{A} = \begin{pmatrix} 0 & 0 & 0 \\ 6 & -6 & -3 \\ -18 & 18 & 9 \end{pmatrix}.$$

Converting to row-echelon form gives

$$\begin{pmatrix} 6 & -6 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

which corresponds to the single equation

$$-6x + 6y + 3z = 0.$$

Solve by back-substitution: $z = c_1$, $y = c_2$, $x = y + z/2 = c_2 + c_1/2$, so

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = c_1 \begin{pmatrix} 1/2 \\ 0 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

so $\begin{pmatrix} 1/2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ form a basis for the eigenspace at 2 . (We were lucky here that the number of basis eigenvectors is as large as the multiplicity of the eigenvalue, so that the eigenspace of 2 was not deficient.)

The exponential solutions built from these eigenvectors are:

$$e^{2t} \begin{pmatrix} 1/2 \\ 0 \\ 1 \end{pmatrix}, \quad e^{2t} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

Eigenspace of $\lambda = -1$: This is the nullspace of

$$\begin{pmatrix} -3 & 0 & 0 \end{pmatrix}$$

$$-\mathbf{I} - \mathbf{A} = \begin{pmatrix} 6 & -9 & -3 \\ -18 & 18 & 6 \end{pmatrix}$$

Converting to row-echelon form gives

$$\begin{pmatrix} 3 & 0 & 0 \\ 0 & -9 & -3 \\ 0 & 0 & 0 \end{pmatrix},$$

which corresponds to the system

$$3x = 0$$

$$9y + 3z = 0.$$

Back-substitution leads to

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = c \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix},$$

so $\begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix}$ by itself is a basis for the eigenspace at -1 .

The exponential solution built from this eigenvector is

$$e^{-t} \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix}.$$

Steps 3. Check whether there are enough independent solutions and write the general solution.

We have three independent solutions,

$$e^{2t} \begin{pmatrix} 1/2 \\ 0 \\ 1 \end{pmatrix}, \quad e^{2t} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad e^{-t} \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix},$$

so they form a basis of all solutions. The general solution is

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = c_1 e^{2t} \begin{pmatrix} 1/2 \\ 0 \\ 1 \end{pmatrix} + c_2 e^{2t} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + c_3 e^{-t} \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix}.$$

If there were initial conditions, we could solve for $\mathbf{c_1}, \mathbf{c_2}, \mathbf{c_3}$ to get a specific solution.

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Problem 5.2

Find the general solution to the system $\dot{\mathbf{x}} = \mathbf{Ax}$ with $\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$. Recall that we have found (in the last lecture) the eigenvalues and eigenspace of $\mathbf{A}$ to be

$$\text{Eigenvalue} \quad \text{Corresponding eigenspace} \tag{4.24}$$

$$\lambda = 0; \text{Span} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \tag{4.25}$$

$$\lambda = 1; \text{Span} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \tag{4.26}$$

$$\lambda = 2; \text{Span} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}. \tag{4.27}$$

Solution

The eigenvalues are all distinct, so we automatically have enough independent eigenvectors. The exponential solutions $\mathbf{ve}^{\lambda t}$ are

$$\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} e^t, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} e^{2t},$$

and the general solution is

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Linearly connected tanks

4/4 points (graded)

You found in a previous problem the system of DE that describes the fluid flow between 3 linearly connected tanks.

Let us consider the simpler case in which 3 tanks are connected (but there is no direct pipe between tanks 1 and 3).

The system of DE is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} \quad \text{where } \mathbf{x} = \begin{pmatrix} h_1 \\ h_2 \\ h_3 \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} -1 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & -1 \end{pmatrix}.$$

(4.29)

The general solution of this system is

$$\mathbf{x}(t) = c_1 \mathbf{v}_1 e^{\lambda_1 t} + c_2 \mathbf{v}_2 e^{\lambda_2 t} + c_3 \mathbf{v}_3 e^{\lambda_3 t}.$$

(4.30)

where $\lambda_1, \lambda_2, \lambda_3$ are scalars that are not necessarily distinct, and $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are vectors in $\mathbb{R}^3$.

Find λ_1, λ_2 , and λ_3 .

(The order of the eigenvalues does not matter.)

$\lambda_1 =$

0

✓ Answer: 0

$\lambda_2 =$

-1

✓ Answer: -1

$\lambda_3 =$

-3

✓ Answer: -3

What is the long term behavior of the system?

☐ The heights of fluid in the three tanks tend to the ratio **1 : 2 : 1**

☐ The heights of fluid in the three tanks tend to the ratio **1 : 0 : 1**

☒ The heights of fluid in the three tanks tend to the same level



Solution:

The unknowns $\lambda_1, \lambda_2, \lambda_3$ and $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ in the given expression for the general solution are the eigenvalues and corresponding eigenvectors of $\mathbf{A}$.

To find the eigenvalues, we need to find the roots of the characteristic polynomial:

$$\det(\lambda \mathbf{I} - \mathbf{A}) = \begin{vmatrix} \lambda + 1 & -1 & 0 \\ -1 & \lambda + 2 & -1 \\ 0 & -1 & \lambda + 1 \end{vmatrix}$$

(4.31)

$$= (\lambda + 1) ((\lambda + 2)(\lambda + 1) - 1) + (-1)(\lambda + 1)$$

(4.32)

$$= (\lambda + 1) ((\lambda + 2)(\lambda + 1) - 2)$$

(4.33)

$$= (\lambda + 1) (\lambda (\lambda + 3)).$$

(4.34)

This gives the eigenvalues: **0, -1, -3**. Since these are all distinct, the corresponding eigenvectors will form a basis of all possible solutions. The general solution is of the form

$$\mathbf{x}(t) = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 e^{-t} + c_3 \mathbf{v}_3 e^{-3t}.$$

(4.35)

Regardless of initial conditions, the solution will tend to the constant solution $c_1 \mathbf{v}_1$.

To find the constant solution $\mathbf{v}_1$, we need to find $\mathbf{NS}(\mathbf{A})$. However, from physical intuition, we know that $h_1 = h_2 = h_3$ is a constant solution, since when the fluid heights are the same in the 3 tanks, there will be no flow.

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Therefore, we can just verify that $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is in **NS(A)**. Indeed:

$$\begin{pmatrix} -1 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

(4.36)

Therefore, the long term behavior predicted by this the solution is that the heights in the 3 tanks will become the same, and this is consistent with physical experience.

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 Answers are displayed within the problem

5. A worked example

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